Self-Directed Project Management Case Study

AI Customer Support Platform - a recruiter-facing summary of the project management approach

Document	16 - PM Case Study
Project	AI Customer Support Platform (fictional company: Kestrel Commerce Ltd.)
Prepared by	Project Manager (portfolio author) - self-directed simulation
Status	Final (simulated project)
Project window	5 Jan - 13 Mar 2026 (simulated project window)

SIMULATED PROJECT - Self-directed portfolio case study. Not professional experience, not client work, not employment. All people, figures and outcomes are fictional.

At a glance

Item	Detail
Type	Self-directed simulated project management case study (not professional work)
Fictional organisation	Kestrel Commerce Ltd. - B2B SaaS for online retailers (about 180 staff)
Product	AI customer support platform: chatbot answering from a knowledge base, human escalation, admin tools, analytics
Duration / method	10 weeks (5 Jan - 13 Mar 2026), Scrum with five two-week sprints
Team (simulated)	8 roles, 61 person-weeks planned capacity
Plan size	46 work packages, 30 user stories (161 story points), 61 requirements, 17 risks, 9 milestones, 15 stakeholders
Non-labour budget (assumption)	5,500 USD including 10% contingency

1. Project overview

Kestrel Commerce, a fictional SaaS company, wanted to answer routine merchant questions instantly using its own help content while keeping humans in the loop for anything complex or sensitive. I planned the project end to end as its project manager in a simulation: charter, scope, requirements, backlog, schedule, risks, stakeholder plan, sprint plan, status reporting and closure. The aim was to show how I combine technical understanding of AI products with project management practice.

2. Business problem

Merchants waited about 6.5 hours on average for answers that already existed in the help center, and about 45% of tickets were routine (baseline assumptions for the simulation). Agents spent their day repeating answers, leaving less time for complex cases. Hiring more agents was slow and expensive. Wrong AI answers on billing or privacy topics could harm trust, so the company needed speed with control.

3. Objectives

Objective	Measure (all are Proposed Project Targets)
Resolve routine questions without an agent	At least 35% of conversations within 8 weeks of pilot go-live
Answer quickly and hand over safely	AI answer in 8 seconds or less (p95); human reply to escalations within 2 business hours
Give correct, sourced answers	Golden-set accuracy of at least 85%; at most 5% unsupported answers; at least 75% positive feedback after launch
Deliver a pilot on time and on budget	Pilot release 11 Mar 2026; non-labour spend within 5,500 USD

4. Users

Five personas shaped the product: a merchant (Customer), a Support Agent, the Support Manager, the Knowledge & Content Lead (Admin) and the Sponsor (Business Stakeholder). Each has goals, pain points and needs in the Project Brief. The personas are fictional; no real users were interviewed.

5. Scope

In scope: grounded AI chat with sources and fallback, human escalation with an agent queue, knowledge-base administration, 30-day conversation history, basic analytics, authentication with four roles, and privacy and security controls. Out of scope: helpdesk/CRM integration, other languages, mobile app or voice, AI actions on accounts and anything beyond the pilot cohort. MoSCoW prioritisation puts 147 of 161 points in Must Have; the 14 Should Have points are a planned de-scope buffer. Scope is protected by a baseline at M2, a published Won't-Have list, one intake route through the Product Owner, impact-assessed change requests and a pre-agreed de-scope order.

6. Solution

Retrieval-augmented generation: the system finds the most relevant knowledge-base passages, asks a language model to answer only from them, shows the sources and falls back to a human when confidence is low. Components: React frontend, FastAPI backend, PostgreSQL with pgvector, an AI layer with PII redaction and guardrails, an LLM/embedding API behind an abstraction layer, an email provider and managed cloud hosting. I chose a simple stack on purpose; the architecture overview explains each part in project-manager language.

7. PM role

In this simulation I acted as project manager. I own the plan and the logs, not the technical work.

The PM does	The PM does not
Plans scope, schedule, capacity and budget; keeps the WBS, timeline, backlog and risk register consistent. Facilitates Scrum ceremonies and removes blockers. Runs risk, issue, dependency and change logs. Prepares status reports and decision requests for the Sponsor. Brings the AI-quality gate and trade-offs to stakeholders early.	Design the UI, write code, tune prompts or run tests. Set product priorities (the Product Owner does). Approve security or privacy sign-off (the Security & Privacy Officer does). Accept the product (the Support Manager does at UAT). Promise accuracy the team cannot evidence.

8. Agile approach

This is a self-directed simulated PM project, and Scrum is the proposed delivery methodology. Scrum fits because time and budget are fixed while AI answer quality is uncertain and must be measured and tuned, so short feedback loops matter. Each sprint ends with a demo of working software and an AI evaluation result.

Sprint	Goal (short)	Points
1 (5 Jan - 16 Jan)	Foundation & Discovery	28
2 (19 Jan - 30 Jan)	Knowledge Base & AI Core	36
3 (2 Feb - 13 Feb)	Customer Chat Experience	34
4 (16 Feb - 27 Feb)	Human Handover, Analytics & Hardening	34
5 (2 Mar - 13 Mar)	Readiness, Release & Closure	29

A Definition of Ready and a Definition of Done (including 'no drop of more than 2 points in AI evaluation') apply. The team is new, so Sprint 1 is lighter and velocity is re-forecast after Sprint 2.

9. Planning

- WBS with 12 areas, 46 work packages and 126 sub-tasks; each work package has a role, dates, dependencies and a deliverable.
- Timeline of 10 weeks with 9 milestones (M1 charter 7 Jan to M9 closure 13 Mar) and a critical path that runs through the AI core, escalation flow and UAT.
- Capacity: 61 person-weeks (305 person-days) against 275 planned person-days, leaving about 10% buffer.

- Backlog: 30 stories of 8 points or fewer, traced to 61 requirements; story sprints match the sprints in which their linked tasks finish.
- Thin schedule slack (about 2 working days) is acknowledged openly; the de-scope order is the contingency.

10. Stakeholder management

Fifteen stakeholders were scored for interest and influence. Four sit in 'Manage closely' (Sponsor, Product Owner, Support Manager and the delivery team); the CTO, Security & Privacy Officer, Legal and regulators (indirect) are 'Keep satisfied'. Examples of tailored engagement: the Sponsor gets a one-page status and decisions with a recommendation; the Security & Privacy Officer reviews the data flow and provider terms early because they can block go-live; agents are involved in usability sessions and UAT to build adoption. The RACI matrix shows one accountable role per activity.

11. Risk management

Seventeen risks are scored on a 5x5 scale with owners, mitigations, contingencies and measurable triggers; five are Critical. The top three are AI hallucination (R01), poor knowledge-base content (R03) and scope creep (R08). Examples: R01 is controlled by grounding, a confidence threshold, a golden-set evaluation each sprint and a go-live gate (at least 85% accuracy, at most 5% unsupported answers); R03 is attacked by a content audit in Weeks 2-5 with named content owners; R08 by change control. In the simulated status report R03 partly materialises (41 duplicate and 52 outdated articles), which shows the register being used, not just filled in.

12. AI-assisted PM workflow

As an AI graduate I use AI as an assistant for first drafts, risk brainstorming, summaries, dependency checks and retrospective clustering. AI is not the decision-maker: I review, validate, prioritise, decide, communicate and approve. Facts and numbers are checked against the logs, and confidential content is not pasted into unapproved tools. See 23_AI_Assisted_PM_Workflow.pdf.

13. Testing and quality

- Test strategy written in Sprint 2: unit and integration tests in CI, golden-set AI evaluation every sprint, security tests (including 20+ prompt-injection cases), load test at 50 concurrent sessions and UAT.
- Quality gates: Definition of Done per story; AI-quality gate before go-live; no open Critical/High defects or vulnerabilities.
- UAT plan with 20 scenarios traced to requirements, run by agents, the Support Manager and the Admin (see 20_UAT_Plan.pdf).

14. Governance

- Steering at every sprint review: Sponsor, CTO, Support Manager, Product Owner and PM.
- Decision rights: Product Owner for priorities, Support Manager for acceptance, Security & Privacy Officer for the security gate, Sponsor for budget and Must-Have scope.
- Change tolerance: Should/Could changes by Product Owner and PM; Must-Have changes, budget above +10% or schedule above +3 working days to the Sponsor.
- One decision log, a weekly risk review, and a status report with a 'decisions required' table.

15. Lessons learned (simulated)

- Open vendor/legal reviews and book SME time in Week 1; they are dependencies the team cannot speed up.
- Put the AI evaluation score into the Definition of Done so regressions surface every sprint.
- A published Won't-Have list makes 'not now' easier to say.
- Plan security-related Should items before the last sprint or they are the first cut.
- Re-forecast from actual velocity after two sprints.

16. Limitations and honest scope of this case study

What is real	What is simulated
The documents, plans, logs, estimates and structure are my own PM work product, drafted with AI assistance and reviewed by me. The plan is internally consistent: IDs, dates, points, dependencies and risks cross-reference.	The company, people, baseline data, team, stakeholder reactions, sprint results, status and closure figures. No software was built, no real users or measurements exist, and no real company, client or employer was involved. Proposed Project Targets are not results.

This case study is not professional experience. It demonstrates how I would plan, structure and communicate a technical project, and where my judgement was applied.